Spec → Code: RID007

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1. Spec

REPORT ID

RID007

EDC SYSTEM / LANGUAGE

veeva / CQL

DESCRIPTION

This report identifies AE overlaps based on matching PT terms and overlapping onset/end dates.

COLUMN HEADING

1. Country
2. Site
3. Subject Number
4. AE PT
5. AE 1 Seq
6. AE 1 Verbatim
7. AE 1 Onset Dt
8. AE 1 End Dt
9. AE 2 Seq
10. AE 2 Verbatim
11. AE 2 Onset Dt
12. AE 2 End Dt

LOGIC

This report lists subjects with overlapping onset dates/end dates for identical AE Terms. Identical AE terms are defined as having the same PT after encoding. Each pair of overlapping AEs should be output on one line.

Note: Continuing AEs (with no End Date) should also be included in the check.

Report should not display deleted records.

If "Randomized only" parameter selection checkbox is checked, the report should only output subjects with Randomization number that is not null.

REPORT SCENARIOS

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SORTING

Default

2. Logic Understanding

This report scans all non-deleted AE records per subject, groups them by matching encoded PT (identical AE term), and flags pairs of AE records within that group whose onset/end date ranges overlap in time - including continuing AEs with no end date treated as open-ended. Each overlapping pair is output as a single row showing subject context plus both AE records' sequence, verbatim, onset, and end date details, optionally filtered to only randomized subjects if the 'Randomized only' parameter is selected.

1. Start with all AE records for each subject, excluding any records marked as deleted, since 'Report should not display deleted records.'

2. Determine the encoded PT (Preferred Term) for each AE record; two AEs are considered to have 'identical AE terms' if they share the same PT after encoding, even if the AE Verbatim text differs.

3. For AE records belonging to the same subject that share the same PT, compare their date ranges (Onset Dt through End Dt) to determine if the two records' active periods overlap in time.

4. When checking for overlap, if an AE record has no End Date (a 'Continuing AE'), it must still be included in the comparison - treat it as ongoing/open-ended rather than excluding it or treating the missing end date as blocking the overlap check.

5. For each pair of AE records (same subject, same PT, overlapping date ranges), output one line representing that pair, populating AE 1 Seq/Verbatim/Onset Dt/End Dt with one record's details and AE 2 Seq/Verbatim/Onset Dt/End Dt with the other record's details.

6. Populate the Country, Site, Subject Number, and AE PT columns for context on each output line, alongside the paired AE 1 and AE 2 details.

7. If the 'Randomized only' parameter checkbox is checked, filter the final output to include only subjects whose Randomization number is not null; if unchecked, include all subjects regardless of randomization status.

8. Since no specific sort order is specified ('Sorting: Default'), output rows in the system's default order rather than a custom-defined sequence - I'm inferring this means no additional sort logic needs to be applied beyond however records naturally come out (e.g., by subject, then AE sequence). *(inferred)*

3. Spec Review

Readiness: Needs revision — The spec's Report Scenarios section is just a copy-paste of the Logic text rather than concrete examples, and the core 'overlap' definition, pairing/de-duplication rules, and PT source field are all left implicit -- these are blocking gaps that would force a query generator to guess at which records and pairs are included.

BLOCKING (completeness): The 'Report Scenarios' field is simply a verbatim copy of the Logic text ('This report lists subjects with overlapping onset dates/end dates...'). There are no actual example scenarios (e.g., 'AE1 onset before AE2 onset, AE1 end after AE2 onset', 'both AEs continuing/no end date', 'AE1 fully contains AE2') showing concrete cases the report should catch or exclude. → Replace the Report Scenarios section with distinct, concrete scenarios (with sample dates) illustrating each type of overlap that should be included, plus at least one non-overlap case that should be excluded, so the generator and testers can verify coverage.

BLOCKING (ambiguity): The Logic never defines what 'overlapping' means mathematically. It's unclear whether overlap requires only date range intersection (AE1.onset <= AE2.end AND AE2.onset <= AE1.end), whether same-day boundary touches count as overlapping, and how open-ended (continuing, no End Date) AEs are compared against AEs that do have an end date. → State the explicit overlap condition, e.g. 'AE1.OnsetDt <= AE2.EndDt (or AE2 is continuing) AND AE2.OnsetDt <= AE1.EndDt (or AE1 is continuing)', and clarify whether equal boundary dates count as overlapping.

BLOCKING (completeness): It's not specified how to avoid duplicate/reciprocal pairs and self-pairing. Since 'each pair of overlapping AEs should be output on one line,' but the spec doesn't state whether AE1 vs AE2 and AE2 vs AE1 (same pair reversed) should both appear, or how to distinguish which AE goes into 'AE 1' vs 'AE 2' columns, or whether a single AE record could overlap with itself (same Seq). → Add explicit rules: exclude self-matches (AE1 Seq <> AE2 Seq), and specify ordering/de-duplication logic (e.g., only output pair where AE1 Seq < AE2 Seq) so each overlapping pair appears exactly once.

WARNING (completeness): Column headings include 'AE PT' (singular, column 4) implying one shared PT value for the pair, but the Logic only says identical AE terms are defined by matching PT after encoding — it doesn't explicitly say this single PT column is populated from either AE1 or AE2's PT (should be same value, but this is only inferable, not stated). → Clarify in the Logic/Column Heading description that 'AE PT' reflects the shared encoded PT term common to both AE1 and AE2, to remove ambiguity for the generator.

WARNING (ambiguity): The report does not define how PT encoding is derived (e.g., MedDRA dictionary field/version) when comparing 'identical AE terms... after encoding.' It's unclear if this refers to a specific coded field on the AE form (e.g., MedDRA PT) or a derived value. → Specify the exact source field (e.g., 'MedDRA Preferred Term field on the AE form, current dictionary version') used for the PT comparison.

WARNING (completeness): The Logic states 'Report should not display deleted records' but doesn't clarify whether this applies only to AE records themselves or also to deleted Subject/Site/Study records, nor what constitutes a 'deleted' record status field. → Specify the field/flag used to identify a deleted record (e.g., a specific status or audit field) and confirm scope (AE-level only vs. subject-level).

INFO (naming): No Form/Item naming convention is stated anywhere in the spec, and Column Headings use generic labels like 'AE 1 Onset Dt' without referencing actual CRF form/item names. → If the organization has a standard Form/Item naming convention, reference the specific AE form name and item OIDs/labels used for Onset Date, End Date, Verbatim Term, PT, and Seq to remove any mapping ambiguity for the generator.

WARNING (testability): There is only one Report Scenario (identical to the Logic), so there's no way to verify distinguishing behavior for edge cases such as continuing AEs, deleted records, or the Randomized-only parameter filter -- the spec doesn't show expected output differences for these cases. → Add scenario-specific test cases (e.g., 'Subject with one continuing AE and one closed AE with same PT and overlapping onset -> included', 'Subject with Randomization Number null and Randomized-only checked -> excluded') to make behavior testable.

4. Generated Code

CQL — Guardrails: passed

```
SELECT a1.`@HDR`.`Site`.`Country` AS `Country`, a1.`@HDR`.`Site`.`Number` AS `Site`,
a1.`@HDR`.`Subject`.`Name` AS `Subject Number`, a1.AESEQ AS `AE/VAE 1 Seq`, a1.AETERM AS `AE/VAE 1
Verbatim`, CODEDTERM(a1.AETERM, 'LLT') AS `AE/VAE LLT`, a1.AESTDAT AS `AE/VAE 1 Onset Dt`,
a1.AEENDAT AS `AE/VAE 1 End Dt`, a2.AESEQ AS `AE/VAE 2 Seq`, a2.AETERM AS `AE/VAE 2 Verbatim`,
a2.AESTDAT AS `AE/VAE 2 Onset Dt`, a2.AEENDAT AS `AE/VAE 2 End Dt`
FROM AE a1
JOIN AE a2 ON a1.`@HDR`.`Subject`.`ID` = a2.`@HDR`.`Subject`.`ID` AND a1.AESEQ < a2.AESEQ
WHERE `@Form`.`Status` = 'submitted__v' AND CODEDTERM(a1.AETERM, 'LLT') = CODEDTERM(a2.AETERM,
'LLT') AND a1.AESTDAT <= IFNULL(a2.AEENDAT, a1.AESTDAT) AND a2.AESTDAT <= IFNULL(a1.AEENDAT,
a2.AESTDAT)
```

Matched deterministic Tier 1 rule 'ae_vae_overlap_self_join' -- no Claude call made. Renders a self-join on AE via two aliased subqueries (a1, a2), matching on CODEDTERM(...,'LLT') and using IFNULL to treat a NULL end date as still-ongoing for overlap purposes. Pattern verified against precedent ABC002_JR; if this spec's Form/Item names differ from standard AE/VAE conventions, verify before running.